



# AI Snapshots

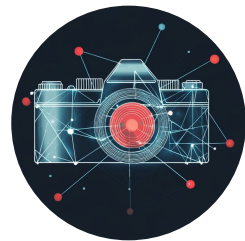
40 warmups for core classrooms

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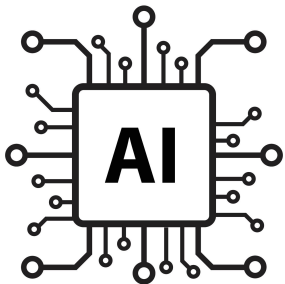
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# What are AI Snapshots?

AI Snapshots are a set of classroom warm-ups designed to help your students build a basic understanding of AI. In only **10 minutes of class time**, students will learn to define, identify, and think critically about artificial intelligence.



**AI Snapshots**



## Why AI?

In the next few decades, AI will affect **any career field** your students pursue, **any consumer product** they use, and **policies** that will shape their lives. In fact, you can bet your students are already interacting with AI in their day-to-day lives. A daily dose of AI will **empower your students** with the knowledge they need to succeed.

# What are AI Snapshots?

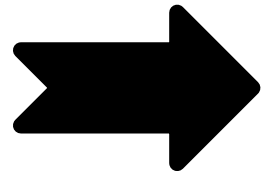


## AI in Core Subjects

Slides 12–16 act as a quick introduction to AI and machine learning. Beyond the introductory warm-ups, real-world examples of AI are considered in the context of each of the four core subject areas. Each slide is labeled with the subject area to which it relates.

## Beyond Specialization

Despite being organized by subject, none of the warm-ups require specialized subject-matter knowledge. We strongly encourage you to do as many warm-ups as you can even if they fall outside of the subject you teach.



# How to use AI Snapshots in your classroom

## 1 Start with the two intro warmups

These baseline snapshots help learners develop a common definition of artificial intelligence.

**1** → *"Smart" machines*

**2** → *At your service*

Henn Na Hotel **replaced 243 staff members with AI-powered robots...** and it didn't go well. **Check-in robots** couldn't answer questions or scan IDs. **Bellhop robots** ran into walls. **Concierge robots** woke guests every time they snored because they thought the guests needed help.

**?** How would you **teach a robot to perform one of these tasks** more effectively?

Introduction to AI

AI Snapshots

## 2 Choose your subjects

Each subject has 10 warmups. Use every warmup or pick and choose your favorites!

**Math warmups**

AI Snapshots for the math classroom

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AI S

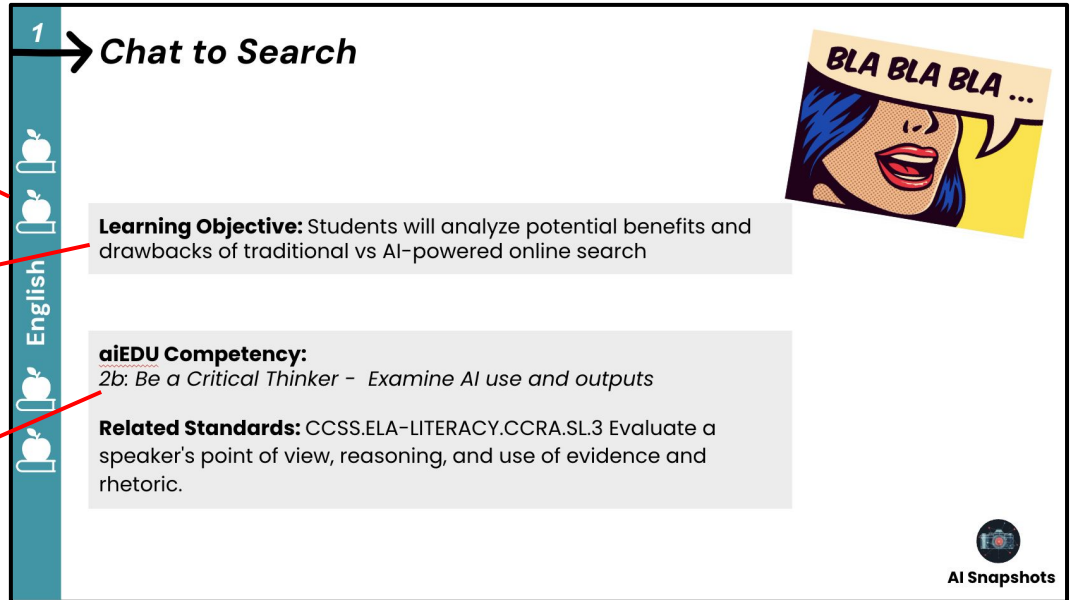
# Anatomy of an AI Snapshot

## Standards and Objectives

The colored bar on the left indicates content area. Icons show this is a **teacher-facing** slide.

AI Snapshots have clear statements of what students should be able to do or understand.

Each AI Snapshot is aligned to an aiEDU AI Literacy Competency and a subject-specific practice standard\*.



The diagram illustrates the structure of an AI Snapshot. On the left, a vertical blue bar contains a white '1' at the top, followed by a right-pointing arrow, and then three icons: a book, an apple, and a speech bubble. The word 'English' is written vertically next to these icons. Three red lines point from the explanatory text on the left to the bar: one to the '1', one to the apple icon, and one to the speech bubble icon. To the right of the bar, the title 'Chat to Search' is displayed. Below the title, there are two gray boxes. The first box contains the text: 'Learning Objective: Students will analyze potential benefits and drawbacks of traditional vs AI-powered online search'. The second box contains the text: 'aiEDU Competency: 2b: Be a Critical Thinker - Examine AI use and outputs' followed by 'Related Standards: CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.' In the top right corner, there is a cartoon illustration of a woman with blue hair and a yellow speech bubble that says 'BLA BLA BLA ...'. In the bottom right corner, there is a small circular logo with a camera icon and the text 'AI Snapshots' below it.

1 → **Chat to Search**

**Learning Objective:** Students will analyze potential benefits and drawbacks of traditional vs AI-powered online search

**aiEDU Competency:**  
*2b: Be a Critical Thinker - Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.

AI Snapshots

\*Common Core Standards for Mathematical Practice, Common Core College and Career Ready Anchor Standards, Next Generation Science and Engineering Practice Standards

# Anatomy of an AI Snapshot

## Activity Page

Each Snapshot begins with a setup or scenario that provides context for the activity.

The main question or prompt that students will respond to. Icons indicate the recommended facilitation strategy.

See the speaker notes for recommended facilitation strategies, key discussion points, student misconceptions, and more.

1

English

### Chat to search

An AI company wants to replace **search engines** with **answer engines**. The new way to search online would involve talking with a chatbot that can answer questions directly instead of a traditional search that returns a list of links.

Proponents say this makes the online search experience easier and more natural, while critics worry about the spread of misinformation and decline in critical thinking.

 Should talking with a chatbot replace traditional search? Why or why not?



**Recommended Facilitation Strategy:** Fist to Five

- Initial Vote (1 minute): Explain the Fist to Five voting system: fist means strongly disagree, 5 fingers means strongly agree. Ask students to show their level of agreement with the statement: "Chatbot online search should replace traditional search."
- Pair Discussion (3 minutes): Students discuss their reasoning with a partner, especially if they had different opinions.
- Class Discussion (5 minutes): Call on students with different vote levels to explain their reasoning. Encourage respectful debate.
- Reconsideration (1 minute): Allow students to change their vote based on the discussion.

# Facilitating Snapshots

These facilitation strategies help **gauge student understanding**, **encourage critical thinking**, and **promote meaningful discussion** of AI concepts.

|                                 |                                                                                                     |                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------|
| Gauging student understanding   |  Fist to Five      | Indicate level of agreement with fingers |
|                                 |  Digital Polls     | Vote using digital tools                 |
|                                 |  Journaling        | Record reflections in journal            |
| Encouraging critical thinking   |  Notice and Wonder | Observe and ask meaningful questions     |
|                                 |  Rank Order        | Rank by order of importance or relevance |
| Promoting meaningful discussion |  Four Corners      | Argue from different perspectives        |
|                                 |  Think-Pair-Share  | Share and respond to others' thinking    |
|                                 |  Discussion Board  | Post and respond on digital forum        |

# Differentiating Snapshots

If students need support **engaging in the activity**, try these:

## **Recruit student interest**

Allow students to choose how they participate (in writing, verbally, or by drawing)

Involve students in setting their own goals for AI literacy development

Relate activities to students' lives and personal experiences

## **Support persistence**

Clearly state and display the Snapshot's learning goal

Use sentence starters to help students talk about AI concepts

Provide frequent, timely, specific feedback in relation to standard or competency

## **Promote self-regulation**

Provide prompts and rubrics to encourage self-reflection

Create opportunities for students to appreciate their strengths and those of others (e.g. sharing notes of appreciation with peers)

# Differentiating Snapshots

If students need support **accessing information**, try these:

## Offer options for displaying information

Offer alternatives to auditory information, like:

- written transcripts,
- diagrams
- speech-to-text tools

Offer alternatives to visual information, like:

- Text or spoken descriptions for image and video
- text-to-speech tools

## Provide options for language and symbols

Pre-teach key vocabulary and symbols.

Make key information available in multilingual learners' first languages.

Make relationships between elements explicit (e.g. highlighting transition words in an essay, links between ideas on a concept map, etc).

## Provide options for comprehension

Provide graphic organizers to organize and link concepts.

Use analogies and metaphors to illustrate concepts.

Chunk information and directions into smaller parts.

# Differentiating Snapshots

If students need support with **demonstrating their understanding**, try these:

## Vary options for interaction

Offer options for physically responding (e.g. alternatives for marking with pen and pencil, alternatives to mouse control)

Create flexible physical spaces (e.g. flexible seating, lighting, etc)

## Offer options for expression and communication

Allow students to compose in multiple media, such as text, speech, drawing, illustration, video, etc

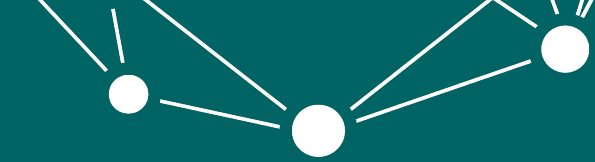
Provide sentence starters, outlining tools, story webs, manipulatives, or web apps to support students in crafting their responses.

## Provide options for strategy development

Use prompts for students to show and explain their work.

Use checklists and guides for note-taking.

Prompt students to identify the types of feedback they are seeking.



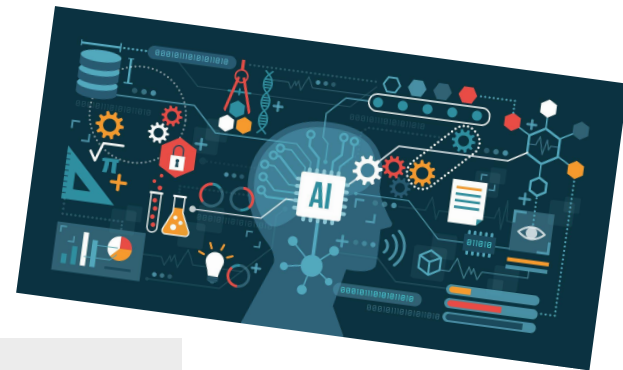
# Two intro warmups

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AI Snapshots for all classrooms



# 1 → "Smart" machines



**Learning Objective:** Students will be able to describe what artificial intelligence is in simple terms.

**aiEDU Competency:**

*1a: Know Your Basics – Know what AI is and how to use it*

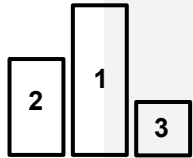
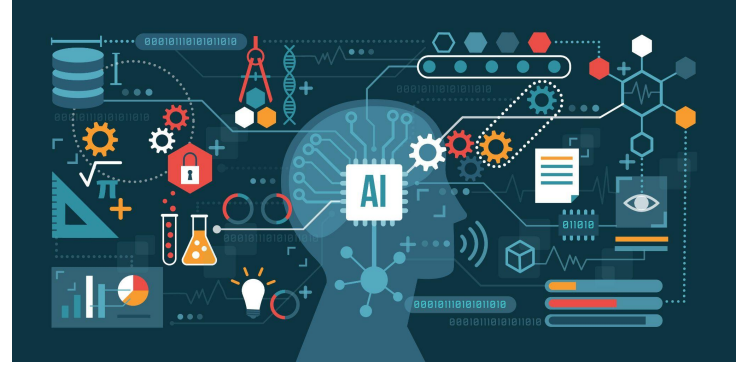
**Related Standards: Related Standards:**

CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



# 1 → "Smart" machines

**Artificial intelligence** is behind much of the technology you use on a daily basis. AI powers the **GPS** that gets you to your destination, knows which **TikTok videos** will make you laugh, and **unlocks your smartphone** with nothing but your face.



If you had to **describe AI to an 8 year old**, what would you say?



## ➔ At your service

**Learning Objective:** Students will think critically about the challenges AI faces in performing human-like tasks.



### **aiEDU Competency:**

*1b: Know Your Basics – Know how AI works*

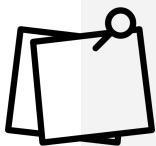
**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## 2 → At your service

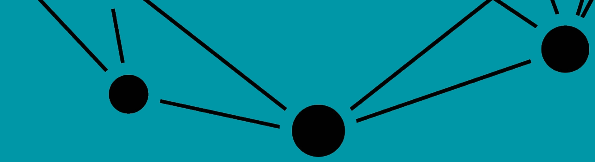
Henn Na Hotel **replaced 243 staff members with AI-powered robots...** and it didn't go well.

**Check-in robots** couldn't answer questions or scan IDs. **Bellhop robots** ran into walls. **Concierge robots** woke guests every time they snored because they thought the guests needed help.



How would you **teach a robot to perform one of these tasks** more effectively?

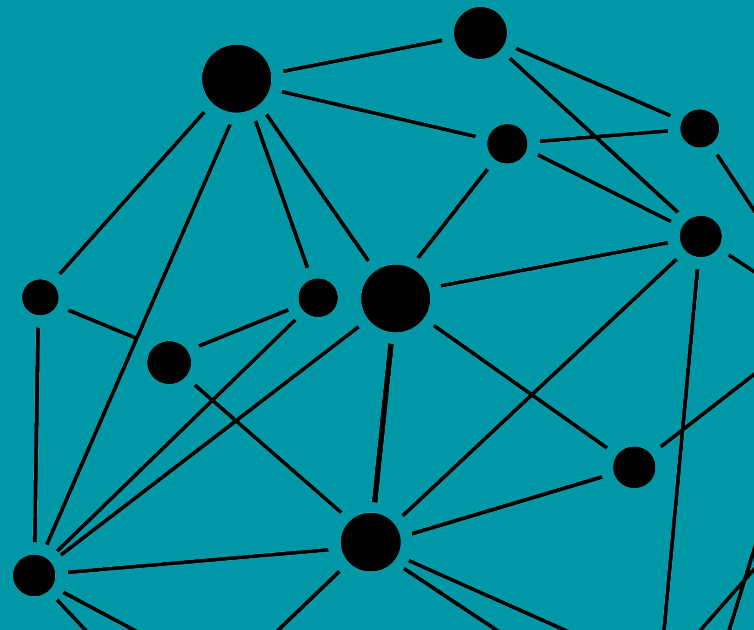




# English warmups

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AI Snapshots for the English classroom



# 1 → Chat to Search



**Learning Objective:** Students will analyze potential benefits and drawbacks of traditional vs AI-powered online search

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



# 1 → Chat to Search

Now there are multiple AI companies that want to replace **search engines** with **answer engines**. The new way to search online would involve talking with a chatbot that can answer questions directly instead of a traditional search that returns a list of links.

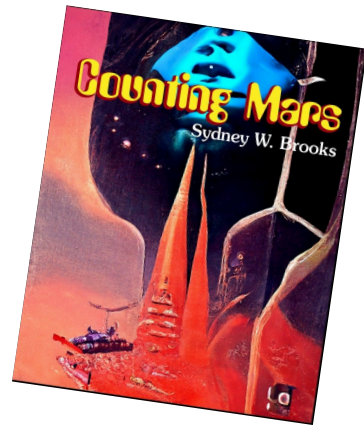
Proponents say this makes the online search experience easier and more natural, while critics worry about the spread of misinformation and decline in critical thinking.



Should talking with a chatbot replace traditional search? Why or why not?



## 2 → Counting Mars



**Learning Objective:** Students will analyze the benefits and harms of using AI to generate book covers.

**aiEDU Competency:**

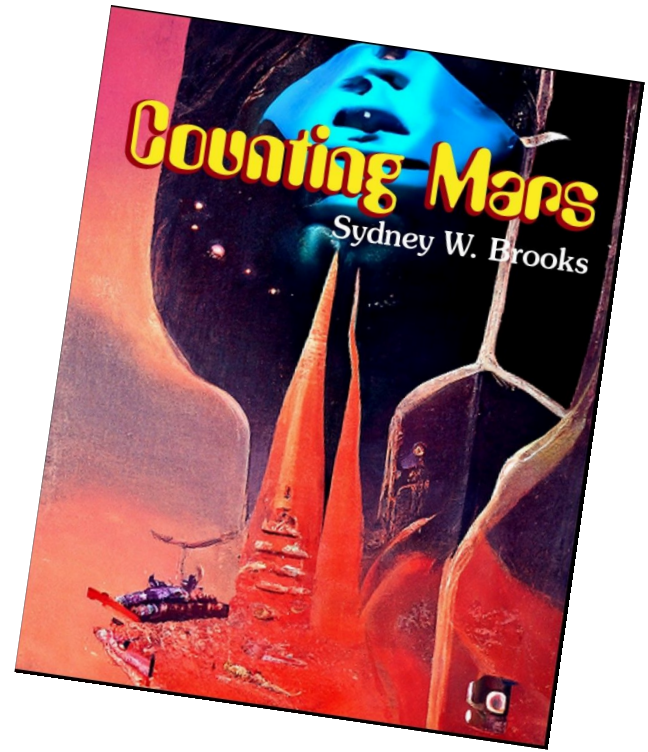
*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## Counting Mars

Some artists are using AI to generate high quality sci-fi novel covers in a tiny fraction of the time it usually takes.



Who could **benefit** from the use of this technology and who could be **harmed** by it?



### 3 → AI Book Critics



**Learning Objective:** Students will evaluate arguments for and against AI-generated book reviews.

**aiEDU Competency:**

*2a: Be a Critical Thinker – Spot and question biases*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



### 3 → *AI Book Critics*

#### Argument A

"Developing AI to write book reviews would help people make better choices. Not only would an AI have a neutral point of view, but it would write reviews faster, too. No more waiting weeks after a book's release."

#### Argument B

"An AI could never appreciate a good book or recognize a bad one the way a real person could. It's important to understand the human perspective when deciding what to read next. We need real live book critics."



Which **argument** do you **agree with more**? **Why**?



→ *I didn't write that...*



**Learning Objective:** Students will explore the positive and negative impacts of AI-generated handwriting technology.

**aiEDU Competency:**

*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.W.1 Write arguments to support claims in an analysis of substantive topics or texts using valid reasoning and relevant and sufficient evidence.

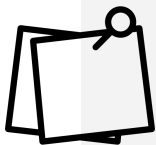


→ *I didn't write that...*

You **forgot to write your name** on that essay you turned in, but it's a good thing your teacher **recognized your handwriting**. A new AI called Hemingway taps into the same skill, but can also **mimic your handwriting style** down to each dotted i and crossed t.



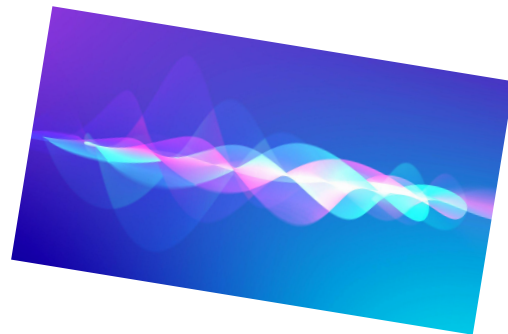
HANDWRITING  
RECOGNITION



Who could **benefit** from this technology and who could be **harmed** by it?



## 5 → Sounds like...me!



**Learning Objective:** Students will analyze the potential impacts of AI-generated voice clone technology on various stakeholders.

**aiEDU Competency:**

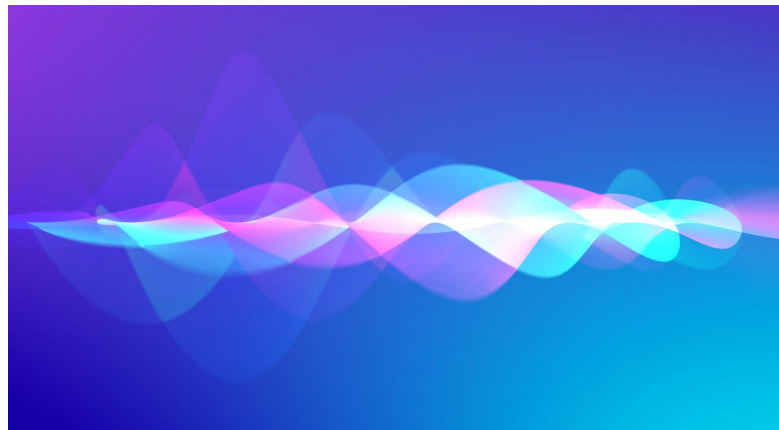
*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.W.1 Write arguments to support claims in an analysis of substantive topics or texts using valid reasoning and relevant and sufficient evidence.



## → Sounds like...me!

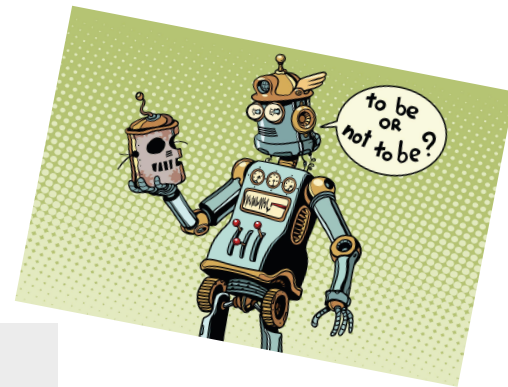
A company called ElevenLabs uses **AI** to learn a user's **unique voiceprint**. The tool uses this information to **create a synthetic copy of a user's voice**. These voice clones can be made to say anything, and are almost indistinguishable from the original.



What are some ways this tool could be used? Would there be any risks to consider?



# → *Methink'st thou art a general offense*



**Learning Objective:** Students will analyze the potential benefits and risks of AI technology that can mimic specific writing styles.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

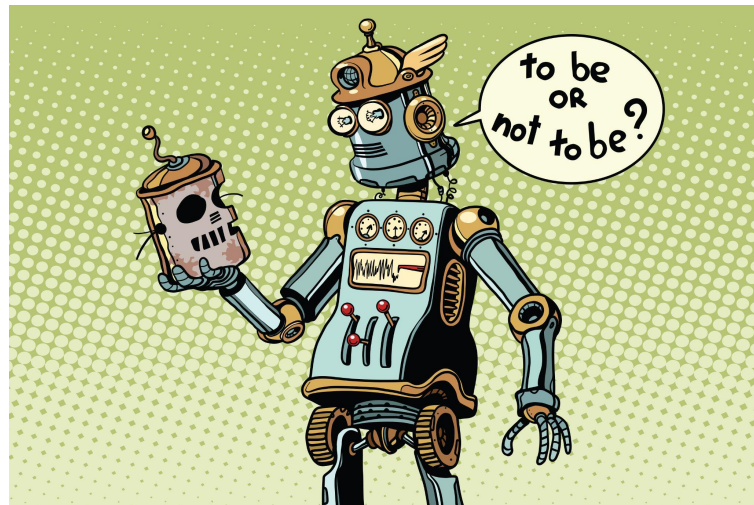
**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4

Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## → *Methink'st thou art a general offense*

AI tools like ChatGPT, Gemini, and Claude can **mimic a writer's style**. For example, it's possible to feed the AI a **description of someone's hairstyle** and receive a **Shakespearean insult** in return.



How could these tools be used? Should there be any limitations?



# → *Death of a journalist*



**Learning Objective:** Students will evaluate arguments for and against AI automation in journalism.

***aiEDU Competency:***

*2b. Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



## → *Death of a journalist*

### Argument A

"It would be easy to automate a journalist using AI. News reporting is often short and to the point. The meat of journalism is combining facts in just the right way to tell a clear story without overly complex language."

### Argument B

"Journalists are unlikely to be replaced by AI. It's not just about generating text—journalists have to conduct research, cultivate sources, and make sense of conflicting facts to tell a story. Like computers or the internet, AI might be a helpful tool at best."



Which **argument** do you **agree with more**? **Why**?



# → No h8



**Learning Objective:** Students will analyze the implications of using AI for hate speech detection on social media platforms.

**aiEDU Competency:**

*2α: Be a Critical Thinker – Spot and question biases*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.W.1 Write arguments to support claims in an analysis of substantive topics or texts using valid reasoning and relevant and sufficient evidence.



## → No h8

**Hate speech and misinformation** are common in certain corners of the internet, and platforms need to know how to **respond**. Facebook reported that in 2021, **97% of the hate speech content** (text, images, videos) that was **removed** from its website was **detected by an AI**.



Why do you think Facebook **relies so heavily on AI** to **detect hate speech**?



# → Where credit is due



**Learning Objective:** Students will explore questions of authorship and copyright in AI-generated content.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



## → *Where credit is due*

Companies like DeepBeat and Suno offer AI tools that can do things like **generate hip-hop tracks** by **learning from vast databases** of existing music.



Who do you think should **get credit for creating** the new raps: the **DeepBeat AI** or the **authors of each individual line**? Why?



# Will you be my friend?

**Learning Objective:** Students will analyze the potential benefits and risks of developing strong emotional connections with AI companions.



**aiEDU Competency:**

*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.1 Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.



## → *Will you be my friend?*

Instagram, Snapchat, and other social media platforms are beginning to offer AI companions that can serve as digital friends. These AI engage in supportive, encouraging conversations and can provide advice or emotional support.

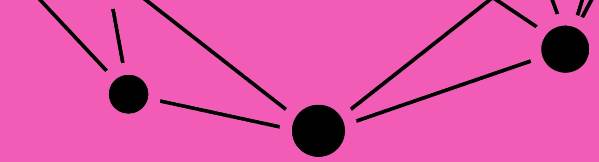


Proponents say AI companions don't replace real friends, but add a new kind of relationship to peoples' lives.



What are the **potential benefits** and **risks** of developing **strong emotional connections** with AI companions?

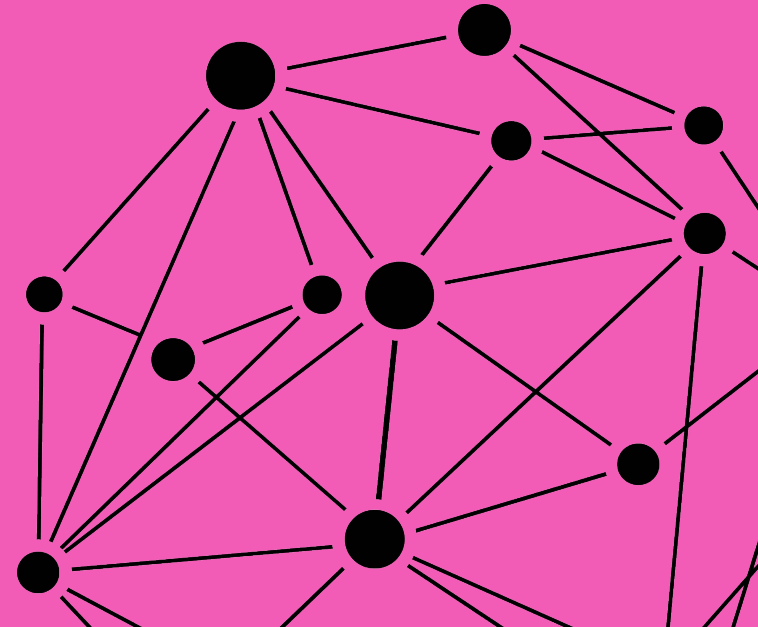




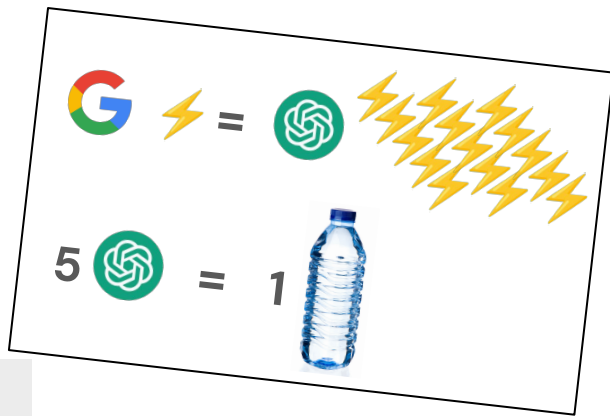
# Science warmups

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AI Snapshots for the science classroom



# 1 → ...but at what cost?



**Learning Objective:** Students will develop questions to evaluate the environmental impact and utility of AI language models.

## **aiEDU Competency:**

*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

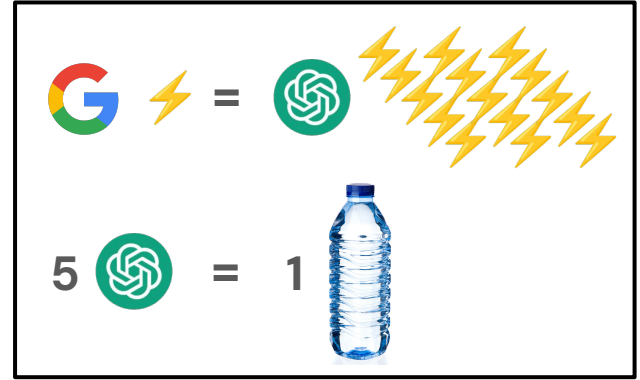
**Related Standards:** NGSS Science and Engineering Practice 1  
– Asking Questions and Defining Problems



# 1 → ...but at what cost?

Large AI models are changing the world, but do you know what it takes to keep them running? Research has shown that:

- a single ChatGPT query uses **15x more electricity** than a single Google query
- 5 ChatGPT queries gulp up about **16 ounces of water** to cool the model's computer systems



What **questions** would you ask about **a new language model** to determine if it was **worth the environmental costs**?



## 2 → Pedal to the metal



**Learning Objective:** Students will evaluate potential real-world applications of an autonomous driving AI system.

**aiEDU Competency:**

*3a – Know the Human Advantage – Be innovative and persistent in pursuit of solutions*

**Related Standards:** NGSS Science and Engineering Practice 6 – Constructing Explanations and Designing Solutions



## 2 → Pedal to the metal

A new AI, dubbed **GT Sophy**, learned to drive by playing the game *Gran Turismo* **millions of times**, getting so good that it **was able to beat some of the world's best players**.



How could this approach be used **in the real world**? If **you were in charge**, **what other games** would you teach AI to play?



→ 3-2-1 ... say honey!



**Learning Objective:** Students will explore innovative applications of AI technology in various fields.

**aiEDU Competency:**

*3b: Know the Human Advantage – Synthesize interdisciplinary and AI knowledge to approach problems holistically*

**Related Standards:** NGSS Science and Engineering Practice 6 – Constructing Explanations and Designing Solutions



## → 3-2-1 ... *say honey!*

Sean Cusak has an interesting resume: in addition to working as an engineer at Microsoft, he's also a **beekeeper**. Recently, Sean created a mini bee photo booth, which **snaps a pic** every time something enters one of his beehives, using AI to **determine** if it was a **bee** or a **pest**.



What other **applications** of this **tool** can you **imagine** beyond identifying beehive pests?



## → Yeah, but why?



**Learning Objective:** Students will evaluate the importance of AI explainability in image classification systems.

**aiEDU Competency:**

*2a: Be a Critical Thinker – Spot and question biases*

**Related Standards:** NGSS Science and Engineering Practice 7 – Engaging in Argument from Evidence



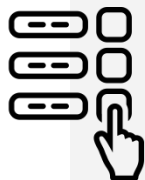
## → Yeah, but why?

### Argument A

"An AI should be able to explain why it labeled images of animals as either predator or non-predator."

### Argument B

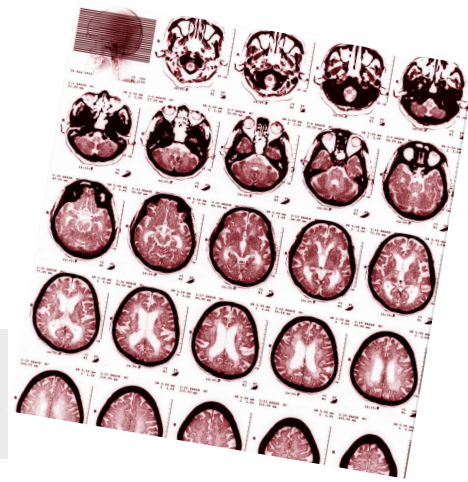
"It doesn't matter if the AI can explain why it labeled images as predator or non-predator as long as it gets them correct."



Which **argument** do you **agree with more** and why?



# → A little of this, a little of that



**Learning Objective:** Students will analyze the potential benefits and challenges of using AI to predict medication responses

**aiEDU Competency:**

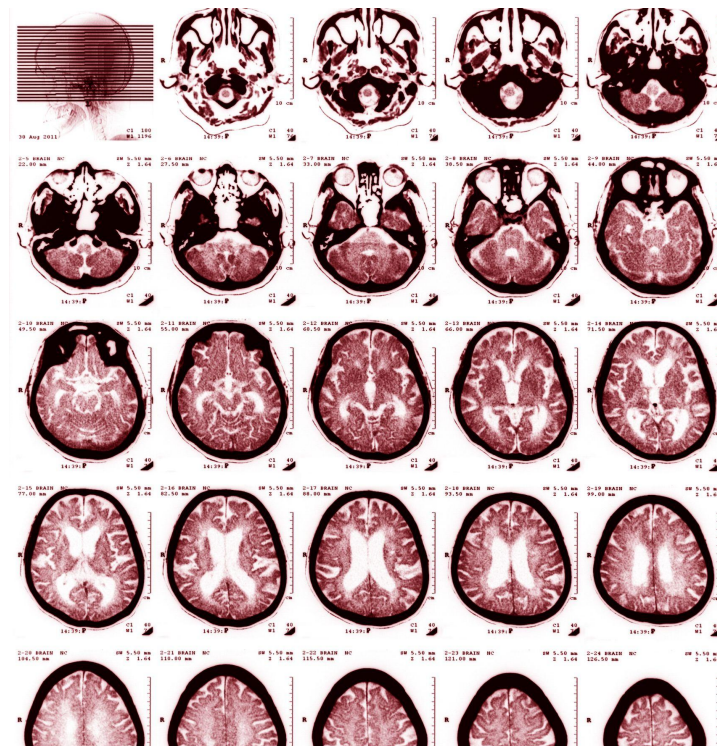
*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** NGSS Science and Engineering Practice 1 – Asking Questions and Defining Problems



## → *A little of this, a little of that*

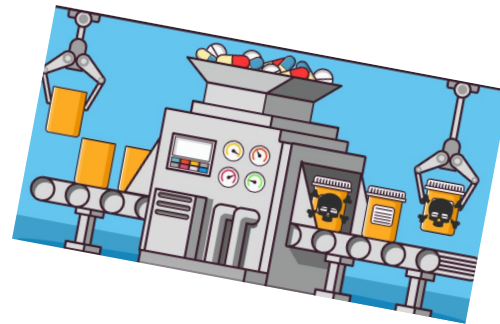
It's common for **people with depression** to **try many different medications** before finding the best one. A new AI tool uses **brain scans** of people with depression to **predict** how well they will **respond to different medications**.



Why is it so **valuable** to have AI that can **predict** how people might respond to different medications?



→ ***I swear it was an accident***



**Learning Objective:** Students will analyze the ethical implications of dual-use AI technology in drug development.

**aiEDU Competency:**

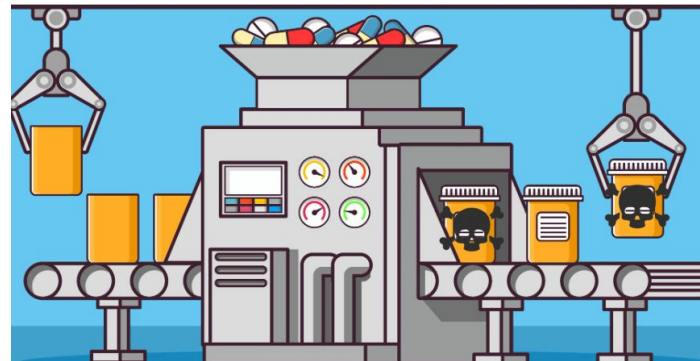
*2a: Be a Critical Thinker – Spot and question biases*

**Related Standards:** NGSS Science and Engineering Practice 1 – Asking Questions and Defining Problems



## *I swear it was an accident*

Researchers in Switzerland developed an AI technology to create new drugs for the **healthcare industry**. The technology would generate new **non-toxic compounds** to treat **human diseases**. They discovered that they could **modify** their technology slightly to **generate poisons** which could be used as **chemical weapons**.



**You're in charge.** How would you **respond** to the development of this new technology?



# 7 → *Beyond the blaze*



**Learning Objective:** Students will explore potential uses of AI technology in various fields.

**aiEDU Competency:**

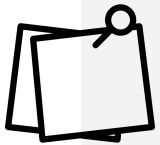
*3b - Know the Human Advantage - Synthesize interdisciplinary and AI knowledge to approach problems holistically*

**Related Standards:** NGSS Science and Engineering Practice 1 - Asking Questions and Defining Problems



## 7 → *Beyond the blaze*

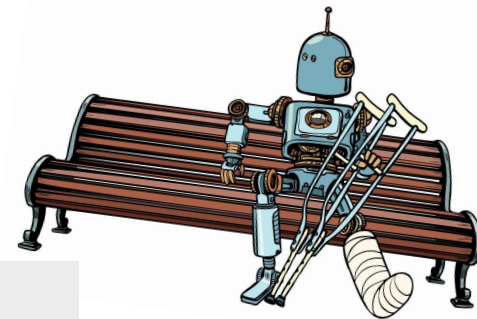
Fighting wildfires is **dangerous** work, especially when the scope of a fire is unknown. In the past decade, the US government has developed **AI-powered drones** to **map** out fires, determine their **direction**, find a fire's **hottest spots**, and set **preventative fires** to destroy wildfire fuel.



Where else do you think this kind of **drone technology** might be **useful**?



# Ouch.



**Learning Objective:** Students will consider the implications of robots with pain-like sensations.

**aiEDU Competency:**

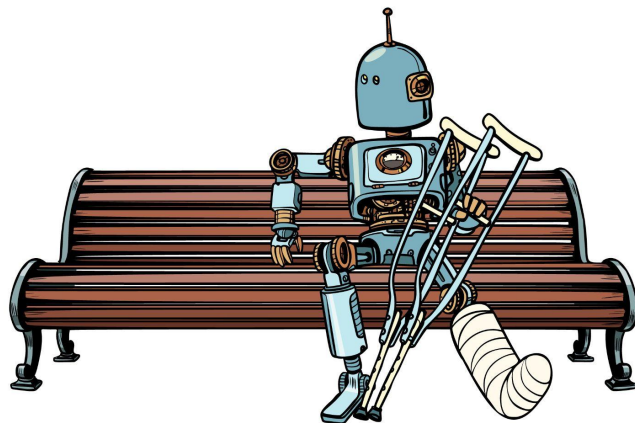
*2a: Be a Critical Thinker - Spot and question biases*

**Related Standards:** NGSS Science and Engineering Practice 1 - Asking Questions and Defining Problems



## → Ouch.

**Pain:** it's the difference between a gentle touch and a bash across the head. A new AI-powered robot developed in Japan can **react to sensations that would cause pain** in humans, complete with human-like facial expressions.



Should robots should be able to **react to a sensation that would cause pain?**



9 → ***\*sniff\* \*sniff\****

**Learning Objective:** Students will explore the potential applications of AI scent recognition technology.

**aiEDU Competency:**

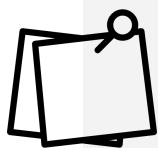
*3a - Know the Human Advantage - Be innovative and persistent in pursuit of solutions*

**Related Standards:** NGSS Science and Engineering Practice 1 - Asking Questions and Defining Problems



## 9 → *\*sniff\* \*sniff\**

Smell is **complex**, but you may be surprised to learn that AI can **learn to recognize scents**. A new AI mimics the brain structures mammals use to smell and can **detect 10 different scents**, such as ammonia. It was **relatively accurate** even when multiple scents were **mixed together**.



**You're in charge. Where** would you **implement** this new technology first to get the **greatest benefit** from it?



# → Paging Dr. Doolittle



**Learning Objective:** Students will evaluate potential uses and ethical considerations of AI models that translate animal communication into human language.

**aiEDU Competency:**

*3a: Know the Human Advantage – Be innovative and persistent in pursuit of solutions*

**Related Standards:** NGSS Science and Engineering Practice 1 – Asking Questions and Defining Problems



## → *Paging Dr. Doolittle*

Researchers are using AI to translate animal communication, from elephant "names" to bat vocalizations. Some scientists aim to develop AI models that would allow humans to speak to animals.



**You're in charge.** Where would you implement this new technology first to get the **greatest benefit** from it?

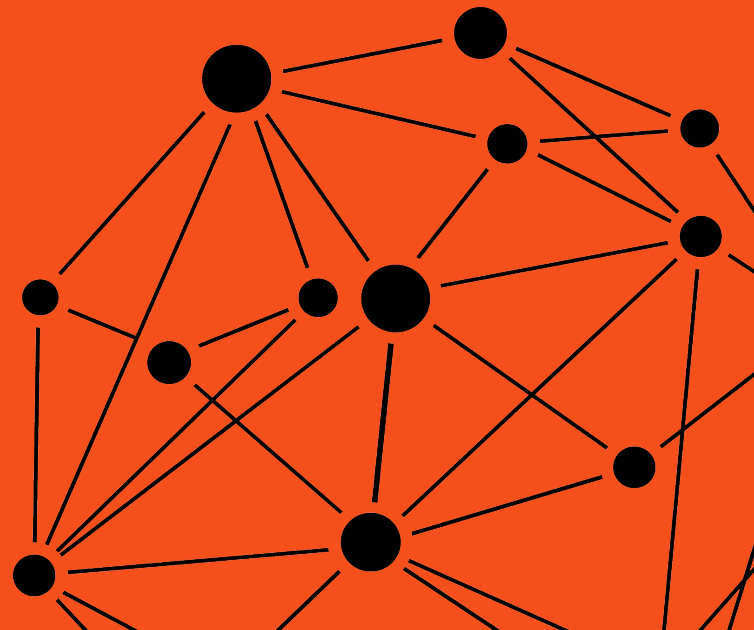




# Math warmups

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AI Snapshots for the math classroom



1

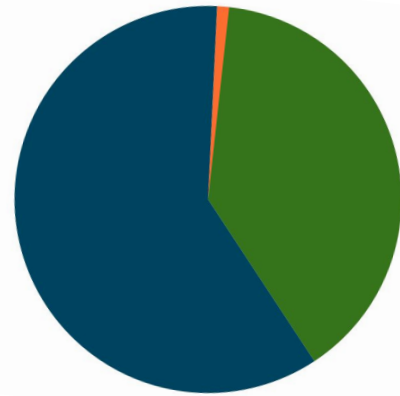
# → Your AI Doctor will see you now...

**Learning Objective:** Students will interpret poll data and consider the implications of AI in healthcare.

**aiEDU Competency:**

*2b: Be a Critical Thinker - Examine AI use and outputs*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.

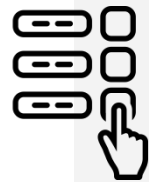
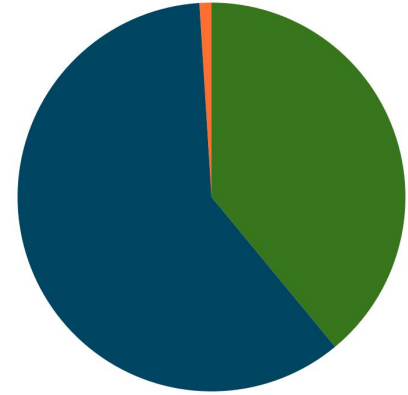


## → Your AI Doctor will see you now...

Researchers have demonstrated that AI is getting very good at diagnosing illnesses, in some cases **performing better than human clinicians**.

However, a **Pew Research Center** survey found that **the majority of people** would **feel uncomfortable** if their healthcare provider **relied on AI** for their medical care.

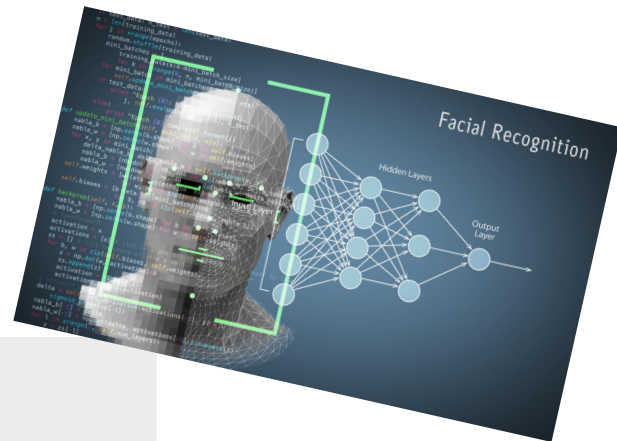
- Comfortable
- Uncomfortable
- No Answer



What **questions** would you ask your doctor in order to **decide whether you are comfortable** with them using AI tools for your own care?



# Do I know you?



**Learning Objective:** Students will evaluate the societal implications of storing and deleting facial recognition data.

**aiEDU Competency:**

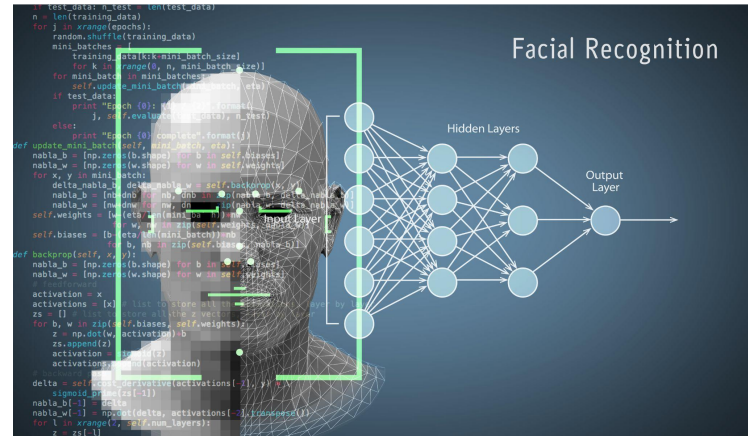
*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



## 2 → Do I know you?

Recognizing a face may seem like a human skill, but AI views it as a **math problem**. In late 2021, Facebook announced it would **delete** its massive store of facial recognition data on **1 billion users**.



Who could **benefit** from **deleting** these templates and who could be **harm**ed?



# AI High

**Learning Objective:** Students will notice patterns and ask questions about data on computer science education distribution in the U.S.

## **aiEDU Competency:**

*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP2 Reason abstractly and quantitatively. Students are asked to interpret percentages and consider their meaning in the context of educational policy and outcomes.

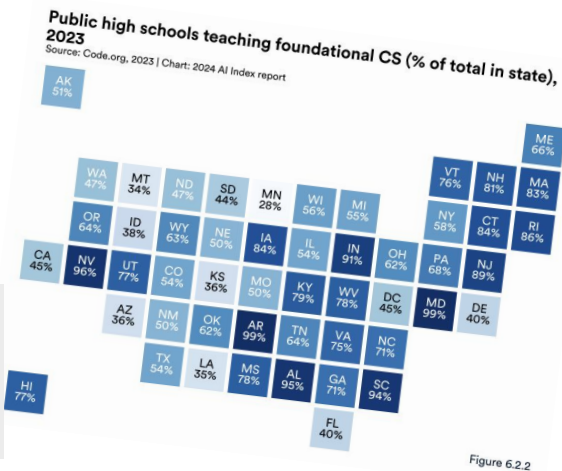


Figure 6.2.2



## Public high schools teaching foundational CS (% of total in state), 2023

Source: Code.org, 2023 | Chart: 2024 AI Index report

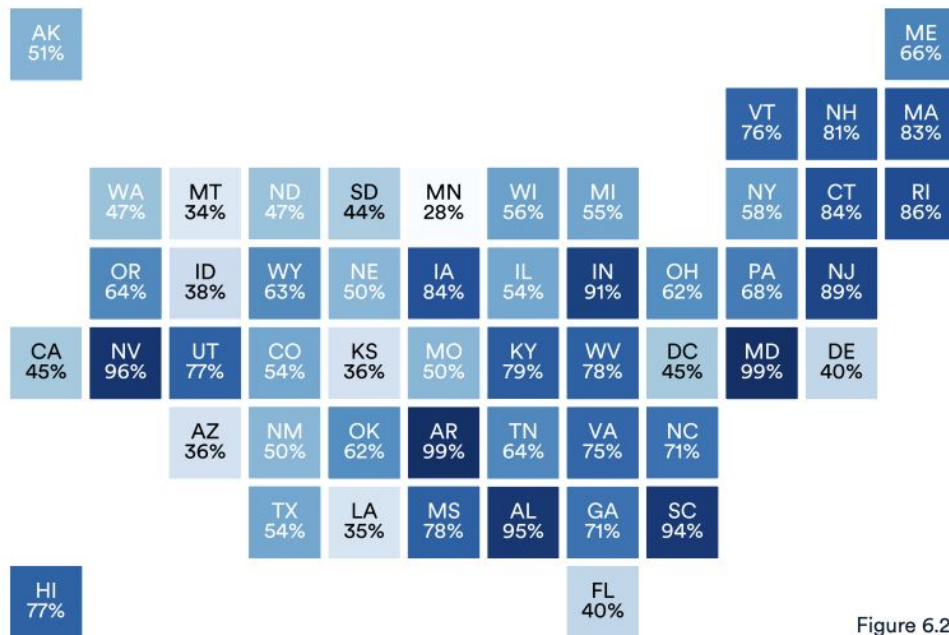


Figure 6.2.2



What do you notice in this chart? What do you wonder?





# It ain't cheap



**Learning Objective:** Students will analyze the costs and potential benefits of training AI models.

## **aiEDU Competency:**

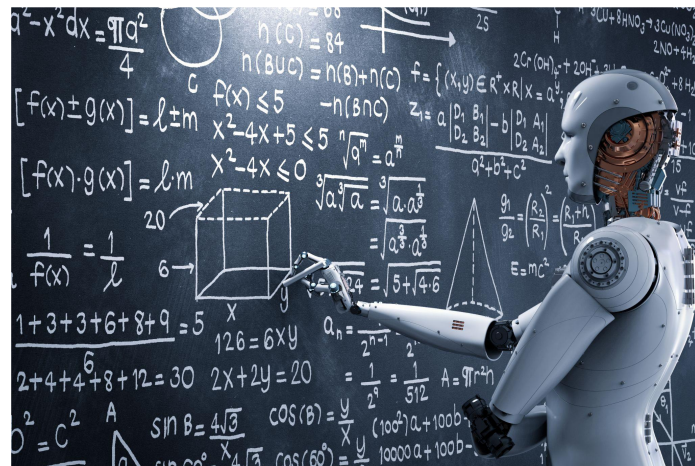
*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



## → It ain't cheap

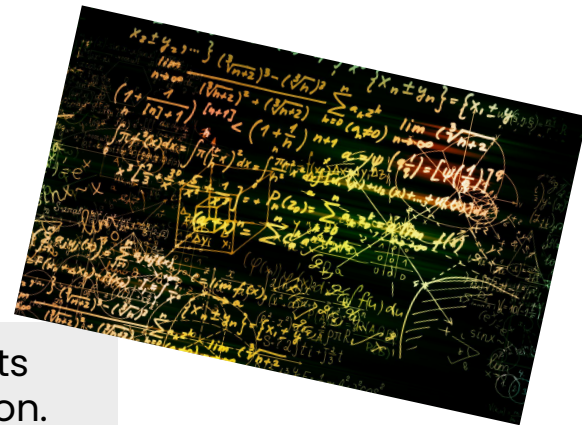
Getting AI to **learn** how to do its job is **pricey** – and it's only **getting pricier**. Smaller models, such as ones used by healthcare providers, might cost **\$50,000**. Larger models, like the one used to power ChatGPT, cost more than **\$100,000,000** to create.



How would you **figure out** whether it's **worth the cost** to **train an AI** for a certain task?



# → An AI that's integral to integrals



**Learning Objective:** Students will analyze the potential benefits and risks of AI technology in advanced mathematics education.

**aiEDU Competency:**

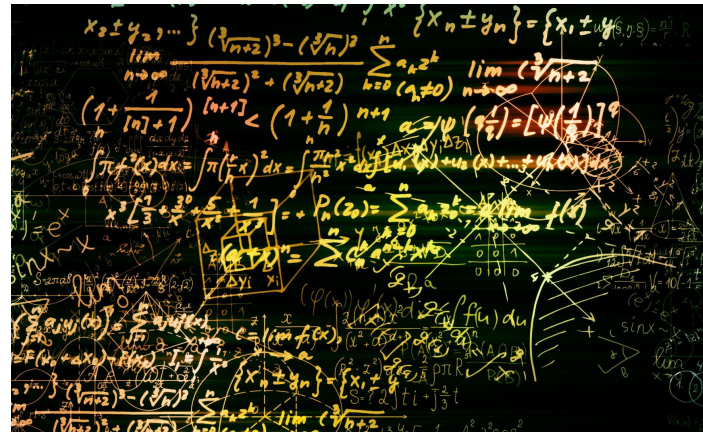
*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



# → An AI that's integral to integrals

Wolfram Alpha has developed a cutting edge AI to **solve calculus problems**. Trained on **tens of millions** of randomly generated problems, the AI can **solve college-level calculus problems**.



What are the potential **benefits** and **risks** of rolling out this technology?



## → Aces and AIs



**Learning Objective:** Students will analyze the potential benefits and harms of AI technology in game strategy.

**aiEDU Competency:**

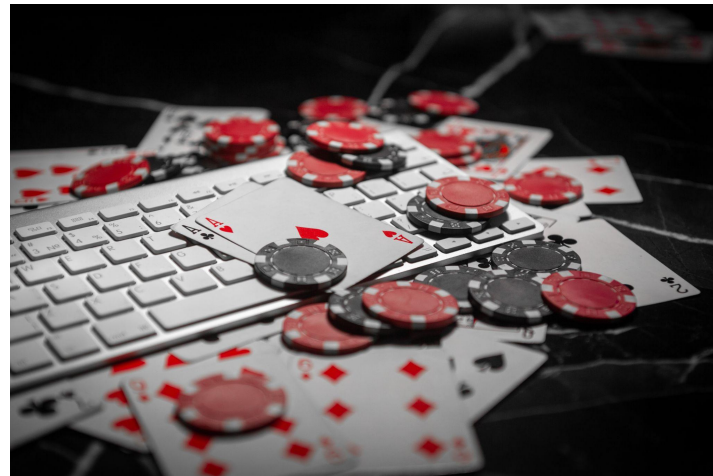
*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



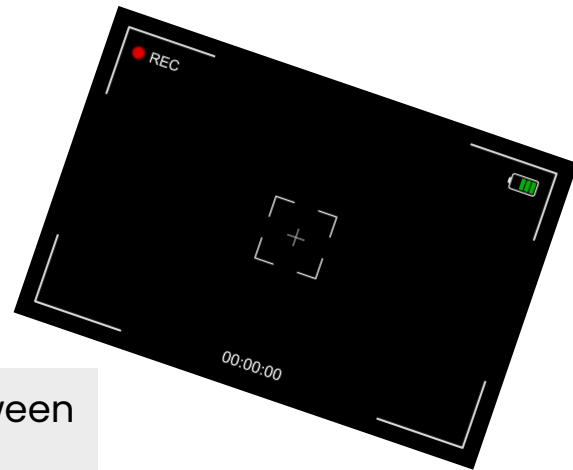
## → Aces & AIs

Betting, checking, and folding in poker are **delicate decisions when money is on the line**. A new AI-based tool called **PioSOLVER** can help poker players train their decision-making abilities by **recommending the best moves in unique situations**.



Why is AI such a powerful tool for helping with decision-making when it comes to playing poker?





**Learning Objective:** Students will analyze the relationship between video length and data available for AI analysis.

**aiEDU Competency:**

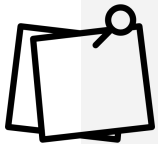
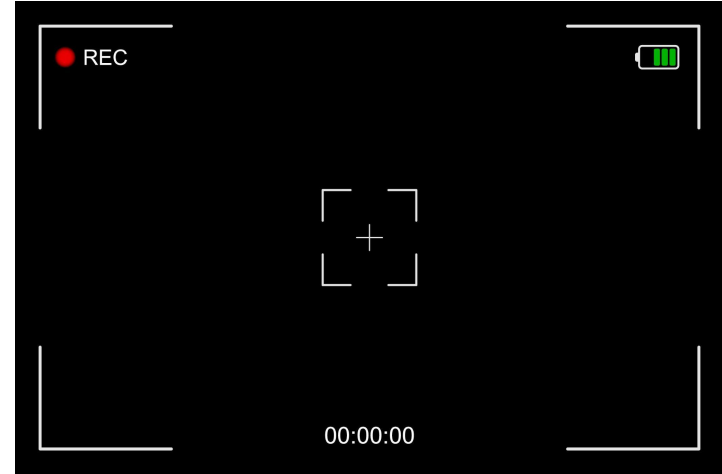
*3a: Know the Human Advantage - Be innovative and persistent in pursuit of solutions*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



## → Action!

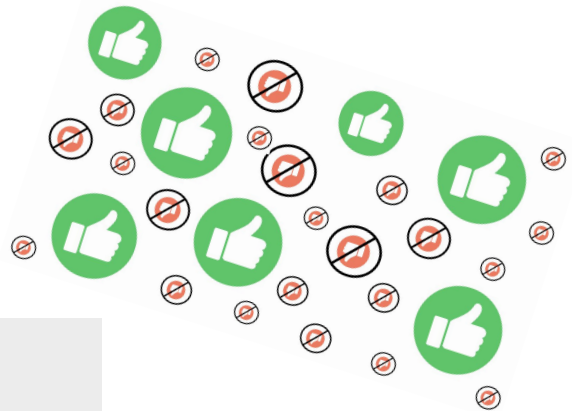
TikTok exploded thanks to their decision to limit videos to **just 15 seconds**. Over time, the company increased the limit to **1 minute**, then **3 minutes**, now **10 minutes**. Observers noted that this might be due to the company's desire to have **more data** for its AI algorithms to analyze.



**You're in charge.** How would you **balance** the benefit of keeping TikTok **distinct** from other video platforms with the **value** of having **more data**?



→ Oh whoa, oh whoa, oh whoa



**Learning Objective:** Students will evaluate the benefits and drawbacks of hiding video engagement data from viewers.

**aiEDU Competency:**

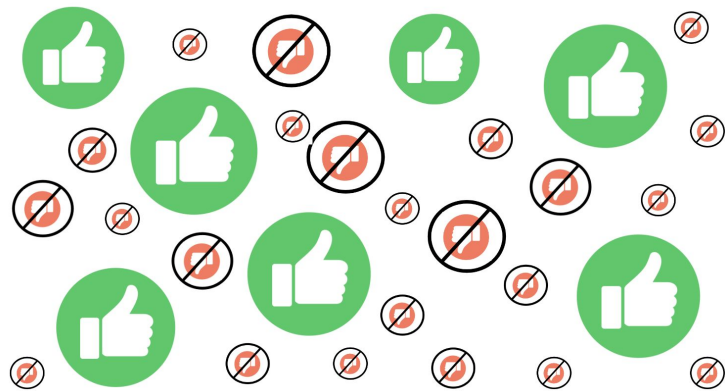
*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



## → Oh whoa, oh whoa, oh whoa

"Baby" by Justin Bieber has over **25 million likes** and **13 million dislikes** on YouTube. This makes it one of the most disliked YouTube videos of all time. But users can **no longer see its like-to-dislike ratio**. While YouTube's algorithm still uses dislikes to rank video suggestions, it **keeps them hidden** from viewers.



What are the **benefits** and **drawbacks** of **hiding dislikes** even though they **still influence suggestions**?



# → Surge Pricing



**Learning Objective:** Students will evaluate the potential impacts of surge pricing on various stakeholders.

**aiEDU Competency:**

*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.MATH.PRACTICE.MP3 Construct viable arguments and critique the reasoning of others.



## → *Surge Pricing*

Rideshare companies like Uber and Lyft have been using AI to raise the price of their rides by **1.1–3 times** in response to variables like high traffic and bad weather. Surge pricing may soon come to other industries. In the future, the price of a cold drink may be higher on a hot summer day.



Should companies be allowed to use AI to **adjust prices** based on **demand** and **weather conditions**?  
**Why or why not?**



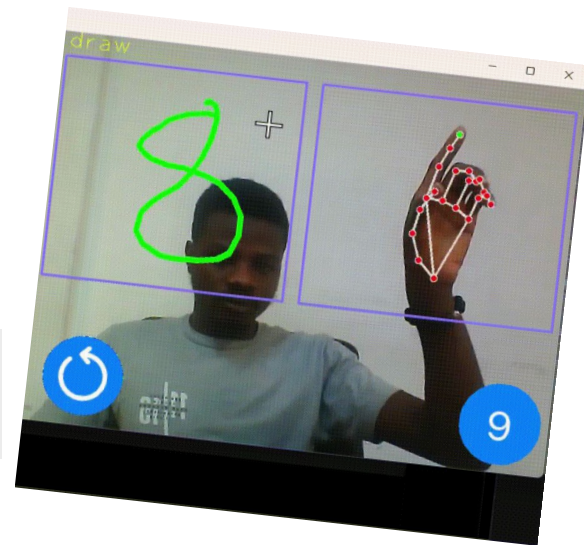
# Digital Revolution

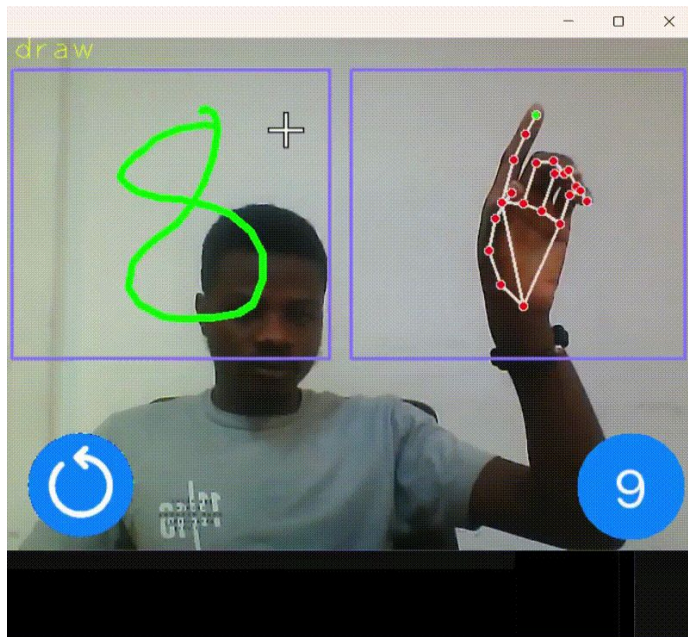
**Learning Objective:** Students will notice and wonder about an AI-powered number recognition system.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

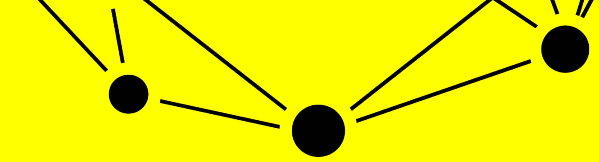
**Related Standards:** CCSS.MATH.PRACTICE.MP1 Make sense of problems and persevere in solving them.





What do you notice? What do you wonder?

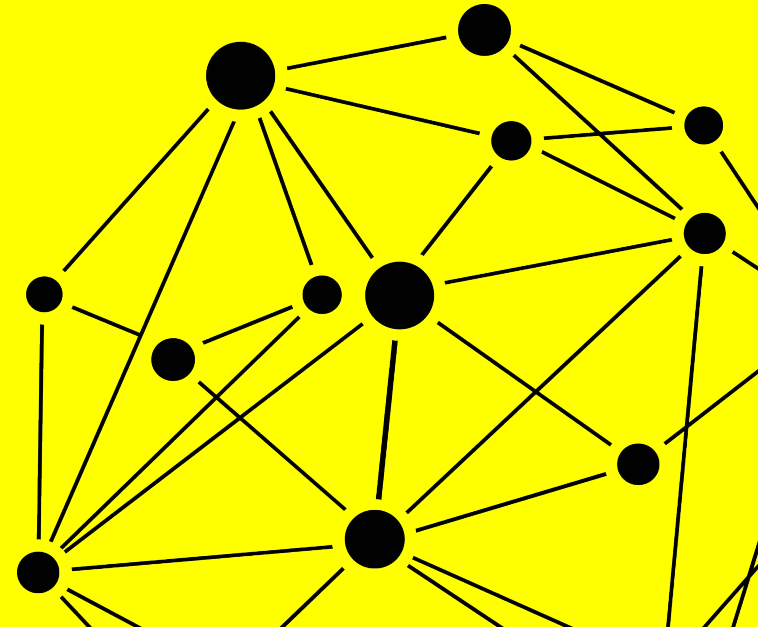




# Social studies warmups

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AI Snapshots for the social studies classroom



# → The buck stops with...

**Learning Objective:** Students will evaluate arguments about the responsibility for regulating unethical AI use in apps.

**aiEDU Competency:**

*2a: Be a Critical Thinker – Spot and question biases*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



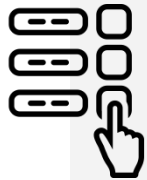
# → *The buck stops with...*

## Argument A

"If an app uses AI in an unethical way, companies like Apple and Google should be responsible for taking it down from their app stores."

## Argument B

"If an app uses AI in an unethical way, the government should be responsible for requiring the author of the app to delete it."



Which argument do you **agree with more** and **why**?



## → Out with the old, in with the new

**Learning Objective:** Students will analyze the potential benefits and risks of AI technology in predicting political instability.

**aiEDU Competency:**

*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.1 Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.



## → *Out with the old, in with the new*

A new AI tech called CoupCast **predicts** whether countries are **likely to have a sudden leadership change** soon. The company gathered data about **factors that contributed to coups** in the past to train their AI to make predictions, **successfully predicting 2021 coups in Mali and Chad.**



**You're in charge.** How would you use this tool? Would you have any **concerns** if it were **freely available** to anyone?





**Learning Objective:** Students will identify and evaluate the essential AI knowledge needed to make informed decisions in daily life.

**aiEDU Competency:**

*1a: Know Your Basics – Know what AI is and how to use it*

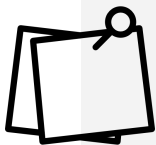
*1b: Know Your Basics – Know how AI works*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.W.1 Write arguments to support claims in an analysis of substantive topics or texts using valid reasoning and relevant and sufficient evidence.



# AI 101

While the US is one of the **strongest AI superpowers**, the **average person doesn't know much about AI**. In a recent study, only **12% of Americans** could pass a basic test about AI.



What do you think **an average person needs to know about AI?**



## Flying solo

**Learning Objective:** Students will analyze the potential benefits and harms of banning autonomous weapons systems.

**aiEDU Competency:**

*2c: Be a Critical Thinker - Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.1 Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.



AI has made its battlefield debut in the war in Ukraine. Should a drone, a gun, or a bomb be able to **decide on its own whether to attack with lethal force**?



How would you respond to a general who is requesting access to the **latest AI weaponry**?



# → Recognizing the face recognition debate



**Learning Objective:** Students will analyze the potential benefits and risks of implementing facial recognition technology in public spaces.

**aiEDU Competency:**

*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.W.1 Write arguments to support claims in an analysis of substantive topics or texts using valid reasoning and relevant and sufficient evidence.



## → *Recognizing the face recognition debate*

In early 2020, the city of London began **rolling out live facial recognition** in some public locations to “scan crowds and identify individuals in real time”. At the same time, the city of Berlin **decided to cancel its effort to install facial recognition** in train stations due to claims that it was **unconstitutional**.



**You're the mayor of your city.** Would you **install facial recognition** technology? **Why or why not?**



# Dating texts (but it's not what you think)



**Learning Objective:** Students will analyze the potential benefits and risks of using AI in historical research and archaeology.

**aiEDU Competency:**

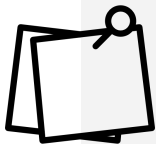
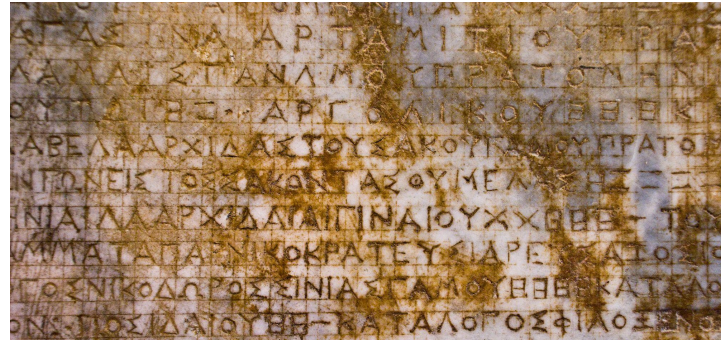
*2c: Be a Critical Thinker – Be continuously curious about AI's influence*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## → Dating texts (but it's not what you think)

An AI called **Ithaca** can “**restore missing text** of damaged inscriptions, **identify their original location**, and help establish the **date they were created** ... Ithaca achieves **62% accuracy** in restoring damaged texts ... and can date texts to **within 30 years** of their ground-truth date ranges.”



**You're an archeologist.** Would you **use** this technology? **Why or why not?**



# → Deepfaking democracy



**Learning Objective:** Students will evaluate the ethical implications of using deepfake technology in political campaigns.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



## → *Deepfaking democracy*

South Korea's recently elected president used **deepfakes** in his campaign:

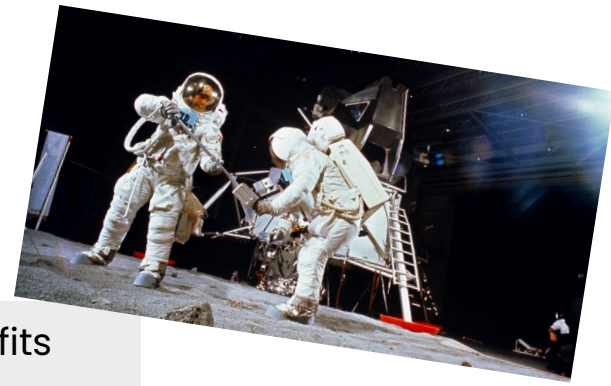
**AI-generated videos of him speaking that appear 100% real.** His avatar was created using 20 hours of video and made him **appear friendlier than usual**, answering questions about his personality type and favorite karaoke songs.



Should politicians be **allowed to use deepfakes** when running for office and **why** or **why not**?



# 8 → AI's truth serum



**Learning Objective:** Students will analyze the potential benefits and drawbacks of implementing AI technology to combat conspiracy theories.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.3 Evaluate a speaker's point of view, reasoning, and use of evidence and rhetoric.



## 8 → AI's truth serum

Researchers have unveiled DebunkBot, an AI chatbot designed to persuade users to stop believing conspiracy theories. A 2024 study has been shown that it can reduce people's false beliefs by 20%.



Photo: The New York Times



Governments and social media platforms should implement this technology. **Do you agree?**



# → Pixelated prejudice

**Learning Objective:** Students will analyze and evaluate the cultural biases present in AI-generated images.

**aiEDU Competency:**

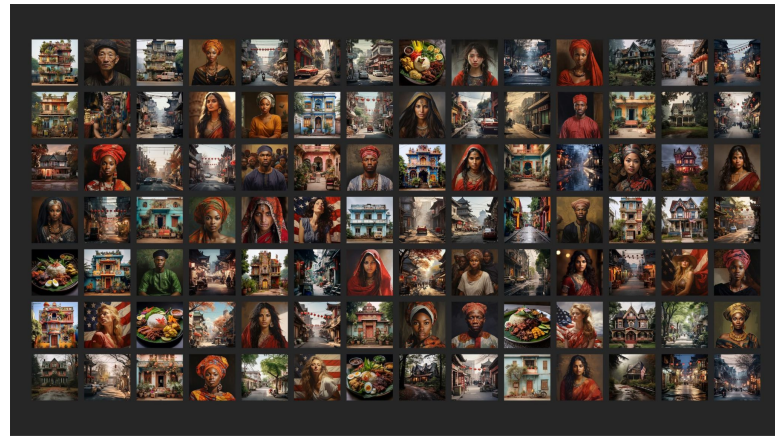
*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-LITERACY.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## → *Pixelated prejudice*

AI image generators like Midjourney can create images based on text prompts. Their results are often **stereotypical or biased** because they rely on datasets that have their own bias.



**Given the likelihood of bias,** what precautions should people take when using this tool?



# → Tag, you're AI

**Learning Objective:** Students will explore the impacts of mandatory tagging of AI-generated content.

**aiEDU Competency:**

*2b: Be a Critical Thinker – Examine AI use and outputs*

**Related Standards:** CCSS.ELA-Literacy.CCRA.SL.4 Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.



## → *Tag, you're AI*

In 2024, China announced plans to require all AI-generated content posted online in China to be clearly labeled. This includes audible disclaimers for audio, watermarks on images and videos, and notices on all posts regardless of format.



Should governments require all AI-generated content to be clearly labeled? Why or why not?

